

Spring & Wagon QC — a quality record for the spring section and the wagon POH line

Runs on one PC in the shop. No internet needed. Every figure on every screen is counted from the record; where the evidence is thin the screen says so instead of guessing. A person decides — always.

1. What it is

Springs	The inspector checks the spring against the strip as today and taps the band — one tap per spring, the six bands of RDSO G-95 Rev-II (Tables 28–33) with their heights on the buttons. The gauge used is on every record; an uncalibrated gauge marks its readings. A crack, corrosion or deformation is condemned by its reason, and the photograph is kept as evidence.
Wagons	Each wagon carries its checklist by category (WMM), wheel readings judged against the drawing's limits, bogie assembly photographs with a blind second pocket count , a parts ledger (off / on / replaced / scrapped, by position), the air-brake test proforma (WMM §720-C) and its stage history from entry to release.
Release	A wagon cannot be released with a condemned spring, a short pocket count or a wheel below the last-shop-issue diameter. Advisories must each be accepted by a named supervisor with a one-time code. The certificate carries a QR that verifies on any phone with no server.
The record	Hash-chained: nothing can be altered after it is written, including by an administrator, and the audit screen proves it on demand. Corrections are recorded as corrections.
The DRM	When today's pile finishes; how many bogies the sorted stock can build and which position limits it; where wagons wait and which one will miss its date; what keeps coming back; the shop's springs against the standard. Every figure carries the count it rests on.

2. What changes on the floor, and what does not

TODAY	WITH THE APPLICATION	WHAT DOES NOT CHANGE
Spring band written in a register at the bench, totalled at the end of the shift	One tap per spring; the day's count, bands and gauge appear as they happen, for the inspector and the supervisor	The strip, the gauge, the inspector's eye and judgement
A condemned spring is a tick mark	A condemned spring has a reason and, where photographing is on, its photograph	Who condemns it
Wheel limits from memory or the drawing in the office	The limit and the drawing number beside the box as the chalk figure is typed	The reading is still taken by hand
Pocket count by one person	Two people count, the second blind; a short count is caught before the exit gate and named on the release checks	The count is a person's count, not a camera's
Release by signature on paper	Release blocked on named defects; advisories accepted by a named supervisor with a one-time code; a certificate that verifies anywhere	The supervisor decides
Monthly figures compiled by hand	Live figures with their counts, from the same record the floor writes	Nothing is estimated or predicted

Where automation is used today — and where it is not

- **Used now:** reading a painted wagon number from the camera (the person confirms it); searching the shipped manuals in their own words with the page cited; a check that flags a reading unlike its neighbours (a question, never a block); a spoken entry read back for confirmation; counting springs in a bogie photograph as an aid to the person counting.
- **Not used for any decision:** no camera classifies a spring; no model produces a figure on any screen. Where a model is allowed to help, it may only pick *which* question to run — the arithmetic is fixed and the rows are shown.
- **Later, only once earned:** the camera may propose a band on its own after blind reads on this shop's own springs have shown it is right often enough. Until then the server refuses it, and the screen says how far it is from earning it.

3. Who can do what

CAPABILITY	INSPECTOR	SUPERVISOR	DRM	ADMINISTRATOR
Record and correct springs; register a wagon; inspect; photograph	✓	✓	—	—
View wagons and their history	own	✓	✓	✓
Release a wagon; override a stage or a band (one-time code, reason recorded)	—	✓	—	—
Stores — issue, restock, reserve	—	✓	—	✓
Audit chain; certificate export; shadow-run log; ask the records	—	✓	✓	✓
Analytics — the DRM dashboards, the standard report	—	—	✓	✓
Approve a learning change; view learning	—	✓	view	✓
Checklist rules; accounts and roles; system settings	—	—	—	✓

Enforced on the server for every request, not by hiding buttons: an address typed by hand is refused. The DRM's role is oversight — it cannot register or inspect a wagon, so the record is always written by the floor. The administrator manages accounts and rules but cannot alter or delete any record.

4. Network — no internet, and what the tablet needs

Nothing leaves the shop

- One PC in the spring section runs everything: the application, the database, the photographs, the manuals index. No cloud, no external service, no internet connection required at any point.
- Tablets reach the PC over the shop Wi-Fi (or one access point in the bay). Each tablet trusts the PC's certificate once; after that the address opens with no warning and the camera and offline mode are allowed.
- Backups are encrypted to a second disk nightly; a restore onto another machine has been drilled and recorded.

When the Wi-Fi drops

- The tablet keeps working. Readings are held on the tablet — the screen says how many are waiting — and send themselves when the network returns. Nothing is lost; nothing is recorded twice.
- Mobile data does not reach the PC: the PC sits on the shop's private network, which the internet has no route to. A tablet on mobile data alone sees nothing; a tablet on Wi-Fi with mobile data on may quietly switch away — so mobile data stays off during work.
- For the DRM at headquarters, a read-only mirror of the nightly backup can be run there; the live system stays in the shop.

5. Photographs — where they go and what they are for

- **Where:** on the shop PC, as files beside the database (`photos/year/month/`), never on the tablet after sending and never outside the shop. Each row of the record keeps the file's SHA-256 hash inside the audit chain, so a photograph altered or replaced on disk is reported as such rather than shown as evidence.
- **What for, today:** the evidence behind a condemned spring or a failed item; the bogie assembly frames that the pocket counts are made from; a wagon's number read from the paint (confirmed by the person).
- **What for, later:** the same photographs, labelled by the people who sorted them, are what the camera would learn from — this shop's own springs, not a dataset from elsewhere. Until that has been earned, they are evidence only.
- **Backup:** carried with the nightly backup, encrypted the same way; restorable file by file, checked against the stored hash.

6. Not yet proven — said before it is asked

The camera on real springs; the pocket count on real bogie photographs; the bearing sound on a real defective CTRB; the shadow-run verdict. All four wait for the shop floor — which is what the pilot is for. Each screen says so where it applies.

7. What it needs, and the proposed start

Hardware

- One Windows PC in the spring section (any ordinary PC of the last five years), on power and the shop Wi-Fi.
- Two Android tablets with rear cameras.
- A second disk or USB drive for the encrypted backup, kept elsewhere.

Running it

- Two files for the shop: START and the backup. Accounts, checklist rules and gauges are managed on the administrator's screen without a programmer.

Proposed start

A **two-week shadow run** in the spring section: one PC, two tablets, two inspectors and one supervisor, recording beside the register as they do today. At the end, the shadow-run screen shows where the two disagreed and who was right. That decides go-live.

Needed to begin

- A PC in the bay with power and Wi-Fi.
- The names of the two inspectors and the supervisor (accounts are created on day one).
- The shop's G-95 Rev-II and gauge register, to check against.